Artificial Intelligence - Detailed Study Material

Artificial Intelligence (AI) is a branch of computer science focused on building systems that can perform tasks requiring human intelligence such as reasoning, learning, problem-solving, perception, and language understanding. Types of AI: 1. Narrow AI – Performs specific tasks. 2. General AI – Human-level intelligence (theoretical). 3. Super AI – Beyond human intelligence (hypothetical). Applications include healthcare, finance, robotics, recommendation systems, self-driving cars, and chatbots.

Machine Learning

Machine Learning (ML) is a subset of AI that enables systems to learn from data without explicit programming. Types of ML: 1. Supervised Learning 2. Unsupervised Learning 3. Reinforcement Learning Common algorithms include Linear Regression, Decision Trees, K-Nearest Neighbors, Support Vector Machines, and Neural Networks.

Deep Learning

Deep Learning uses multi-layer neural networks to model complex patterns in data. Popular architectures: - Convolutional Neural Networks (CNN) - Recurrent Neural Networks (RNN) - Transformers Transformers power modern Large Language Models (LLMs).

Natural Language Processing

Natural Language Processing (NLP) enables machines to understand and generate human language. Tasks include: - Sentiment analysis - Text classification - Question answering - Chatbots LLMs are trained on large datasets to generate context-aware responses.

Retrieval-Augmented Generation (RAG)

RAG combines retrieval systems with language models. Steps: 1. Extract text from PDFs. 2. Split into chunks. 3. Generate embeddings. 4. Store in vector database. 5. Retrieve relevant chunks. 6. Generate grounded answer. RAG improves factual accuracy and reduces hallucination.